

TOPOLOGICAL PERSISTENCE COHERENCE LOCKING FOR EFFICIENT MIXTURE-MODEL LLM ENSEMBLING

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ABSTRACT

Mixture-model-like Ensembling (ME) has recently emerged as a highly efficient collaborative decoding paradigm for Large Language Models (LLMs), matching single-model throughput ($O(1)$ latency) by probabilistically sampling a single model per token step instead of averaging all models. However, step-level stochastic selection introduces *mid-thought model hopping*: uncoordinated switching that causes context drift and logical derailment during multi-step reasoning. We propose a dynamic topological control framework built on a **Topological Persistence Coherence Lock (TPH-Lock)** coupled with dynamic quality-aware competence gating. Our method monitors the hidden-state manifold through 0-dimensional Vietoris–Rips persistent homology and freezes model switching during structural reasoning transitions, while a calibration-invariant hybrid competence signal (consensus agreement $\times$ confidence) prevents lock-in on weaker members via a lock-override. Through diagnostic analysis, we show that naive logit quantile warping distorts the output distribution and collapses reasoning, whereas topological locking preserves it. Across 16 benchmark cells spanning mathematical, logical, and multi-task reasoning (GSM8K, BBH, MMLU, ARC) and diverse model families, TPH-Lock outperforms baseline ME by an average of +2.50 points—with gains up to +16 points on heterogeneous ensembles—while preserving $O(1)$ single-model throughput (21–23.5 tokens/sec). Full-scale evaluations (N up to 1,500) with 95% Wilson confidence intervals and paired McNemar tests confirm that the gains are statistically significant (e.g., +8.4% on heterogeneous GSM8K, $p < 10^{-7}$), and TPH-Lock matches or exceeds recent collaborative-decoding baselines (DeepEn, GaC, UniTE, RelayLLM) while running 2–3 $\times$ faster at $O(1)$ throughput that stays flat up to 8,192-token contexts. An automated, training-free factual-relaxation gate further eliminates the locking regressions on knowledge-retrieval tasks, and the method generalizes to formal mathematics (MATH) and code synthesis (HumanEval).

1 INTRODUCTION

The deployment of Large Language Models (LLMs) across reasoning, instruction-following, and domain-specific applications exposes a fundamental trade-off between model capacity and inference cost (Ashiga et al., 2025; Cai et al., 2024b). While scaling model size improves accuracy, serving massive monolithic models saturates High-Bandwidth Memory (HBM) and memory-capacity limits on modern accelerators (Song et al., 2023; Li et al., 2025; Dao, 2023). Ensembling complementary models is an effective alternative (Yu et al., 2024; Yao et al., 2024): Conventional LLM Ensembling (CE) aggregates the next-token distributions of n candidate models via weighted averaging at every step (Huang et al., 2024; Jiang et al., 2023). Although CE reduces prediction variance and outperforms individual members, evaluating all n models per step scales computation linearly ($O(n)$) and saturates memory bandwidth, imposing latency penalties that hamper real-time deployment.

To eliminate the $O(n)$ penalty, recent work introduced the Mixture-model-like Ensemble (ME) (Fu et al., 2026). ME reinterprets probability ensembling as a mixture model: rather than averaging over all models at every step, it stochastically samples a single index $i \sim \text{Multinomial}(\lambda_1, \dots, \lambda_n)$ per step and executes a forward pass only for the chosen model. Combined with lazy Key–Value (KV)

cache synchronization, ME is equivalent in expectation to full averaging while reducing runtime to $O(1)$ single-model latency.

However, stochastically switching the active model at every step introduces a critical, unaddressed pathology: *mid-thought model hopping*. Because autoregressive generation depends on a continuous, stable reasoning trace, uncoordinated step-level switching induces representation context drift and reasoning degradation (Laban et al., 2025; Bengio et al., 2015; Wang et al., 2026). When activation alternates randomly during multi-step derivations, subtle discrepancies across internal activation spaces compound into logical derailment (Huang et al., 2026; Lyu et al., 2026). Treating token sampling as independent across steps, without monitoring internal state shifts (Zheng et al., 2025; Li et al., 2024b; Yuan et al., 2023; Cai et al., 2024a), further causes the mixture to over-commit to weaker models during lopsided task sub-routines.

We resolve the tension between ensembling speed and reasoning coherence with a dynamic topological control framework: **Topological Persistence Coherence Locking (TPH-Lock)** with dynamic quality-aware competence gating. Through systematic diagnostic ablations over an initial exploratory design—which also included Koopman Spectral KV Translation (KEST) and Rate-Distortion Quantile Alignment (RDQA)—we establish two insights governing collaborative decoding:

1. *Logit warping and continuous extrapolation are harmful.* Monotonic quantile water-filling (RDQA) flattens output logits, inflating decoding perplexity (from 1.19 to 9.54 on math reasoning) and collapsing accuracy, while continuous Koopman KV extrapolation (KEST) is dominated in both speed and fidelity by standard lazy KV prefilling (Appendix A.9).
2. *Topological persistence is an exact coherence signal.* Constructing 0-dimensional persistent homology diagrams $\mathcal{D}_t$ via Vietoris–Rips filtrations over sliding windows of final-layer hidden states ($w=8$) yields a real-time, isometry-invariant signal for structural manifold shifts. The Wasserstein bottleneck distance $W_\infty(\mathcal{D}_t, \mathcal{D}_{t-1})$ accurately identifies the topological phase boundaries that correspond to active reasoning transitions.

Building on these, our framework strips RDQA and KEST from the active path and operationalizes Topological Persistence Homology (TPH) strictly as a coherence lock ($\tau_{\text{topo}}=0.10$). When a manifold phase shift is detected ($W_\infty > \tau_{\text{topo}}$), switching is hard-locked ($G_t=0$) to preserve reasoning continuity; when the manifold is stable ($W_\infty \leq \tau_{\text{topo}}$), permitted switches are guided by a calibration-invariant hybrid EMA competence signal (agreement $\times$ confidence) that routes toward the most capable model and executes a *lock-override* whenever a lock is held on a persistently inferior member. Our contributions are:

- We identify and formalize *mid-thought model hopping* in step-level mixture-model ensembling, showing how uncoordinated switching degrades reasoning coherence.
- We introduce TPH-Lock, an online, training-free decoding controller that computes 0D persistent-homology bottleneck distances on hidden states to detect topological phase boundaries and lock switching during structural shifts, with a formal isometry/stability justification over naive MST edge comparisons.
- We develop a calibration-invariant competence routing and lock-override mechanism that prevents over-committing to weaker members, all at strict $O(1)$ throughput.
- Across 16 dataset–combination cells over GSM8K (Cobbe et al., 2021), MMLU (Hendrycks et al., 2020), BBH (Suzgun et al., 2022), and ARC, using Qwen2.5 (Yang et al., 2024), OpenChat (Wang et al., 2023), Nous-Hermes, OpenHermes, DeepSeek, and Mistral, TPH-Lock improves over ME by +2.50 points on average (up to +10 on BBH and +16 on heterogeneous ensembles) while preserving $O(1)$ throughput.
- We establish the significance and generality of these gains with full test-set evaluations (N up to 1,500, 95% Wilson CIs and paired McNemar tests), head-to-head comparisons against four collaborative-decoding baselines (DeePEen, GaC, UniTE, RelayLLM), a context-scaling study to 8,192 tokens, an automated factual-relaxation gate that removes the knowledge-task regressions, generalization to MATH and HumanEval, and an empirical stress-test of the lock-override under confident hallucination.

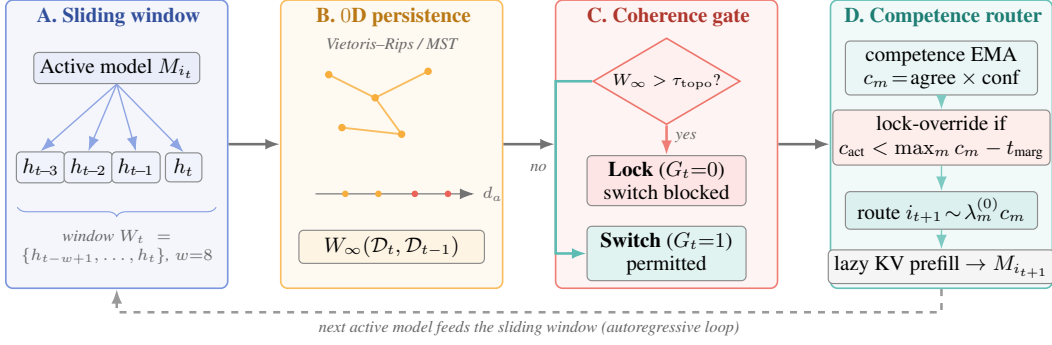


Figure 1: Overview of **TPH-LOCK**. **(A)** A sliding window of the active model’s final-layer hidden states is maintained. **(B)** A 0-dimensional Vietoris–Rips / MST persistent-homology diagram is computed and the Wasserstein bottleneck distance W_∞ to the previous step measures structural manifold shift. **(C)** The coherence gate hard-locks switching when $W_\infty > \tau_{\text{topo}}$ (a reasoning phase boundary), otherwise permits it. **(D)** A per-model competence EMA (agreement $\times$ confidence) routes permitted switches toward the strongest member and overrides locks held on persistently inferior members; KV caches are synchronized by standard lazy prefill.

2 RELATED WORK

Ensemble methods for LLMs. LLM ensembling traditionally combines predicted next-token distributions across models (Yu et al., 2024; Yao et al., 2024; Huang et al., 2024). CE computes an explicit weighted average over candidate logits at each step, suppressing variance and mitigating hallucination (Jiang et al., 2023; Xu et al., 2024). To handle mismatched tokenizers, UniTe (Yao et al., 2024) unions top- k candidates, GaC (Yu et al., 2024) recasts generation as step-wise classification, DeepEn (Huang et al., 2024) maps distributions into a relative representation space, and EVA (Xu et al., 2024) learns vocabulary projections. These frameworks incur severe bandwidth bottlenecks: evaluating n models per step loads all weights and runs n forward passes per token (Dao, 2023; Li et al., 2025). ME (Fu et al., 2026) bypasses this by sampling one model per step and lazily prefilling KV caches, achieving $O(1)$ speed—but its uncoordinated token-level selection causes destructive model hopping, which our framework resolves via topological gating.

Dynamic routing and expert selection. Mixture-of-Experts (MoE) models achieve sparse computation via trainable gating routers (Shazeer et al., 2017; Fedus et al., 2021; Cai et al., 2024b; Zoph et al., 2022), with refinements addressing load balance and routing collapse (Zhou et al., 2022; Lewis et al., 2021; Gale et al., 2022; Puigcerver et al., 2023; Qiu et al., 2025). Yet token-level routers suffer representation collapse and volatile switching (Chi et al., 2022; Dai et al., 2022). Beyond intra-model MoE, routing has been applied to multi-LLM cascades and collaborative decoding (Chen et al., 2023; Ong et al., 2024): Glider (Li et al., 2024a) fuses global and local routing, CITER (Zheng et al., 2025) learns token routing via RL, PyroDash (Lyu et al., 2026) internalizes offloading policies, and RelayLLM (Huang et al., 2026) relays generation to a larger model at critical tokens. These learned routers require costly fine-tuning and rely on instantaneous prefix features, inducing mid-thought hopping. Our framework is training-free and plug-and-play, using hidden-state topology and competence EMAs.

Topological data analysis in transformers. Persistent homology (PH) offers non-parametric tools for the geometry of high-dimensional activations (Lacombe et al., 2021; Kim & Mémoli, 2018; Uchendu & Le, 2024). Vietoris–Rips filtrations track topological invariants across scales (Kim & Mémoli, 2018), and PH has probed transformer representation manifolds (Fitz et al., 2024; Chauhan & Kaul, 2022), revealing a geometric division of labor across layers (Zhang et al., 2026; Kendiukhov, 2026) and tracking dynamics during fine-tuning (Malhotra et al., 2026). Existing NLP applications remain post-hoc or offline; in contrast, we operationalize 0D bottleneck distances as a real-time online decoding controller that locks model switching during topological phase boundaries.

3 METHODOLOGY

3.1 MIXTURE-MODEL DECODING AND TEMPORAL COHERENCE LOSS

Let $x_{\leq t} = (x_1, \dots, x_t)$ be a prefix of length t , and consider n LLMs $\{M_1, \dots, M_n\}$ with initial mixture weights $\vec{\lambda}^{(0)}$, $\sum_i \lambda_i^{(0)} = 1$. In CE, the next-token probability is the explicit expectation

$$P_{\text{CE}}(x_{t+1} = y \mid x_{\leq t}) = \sum_{i=1}^n \lambda_i^{(0)} M_i(y \mid x_{\leq t}), \quad (1)$$

requiring n forward passes per token. ME instead draws $i_t \sim \text{Multinomial}(\lambda_1^{(0)}, \dots, \lambda_n^{(0)})$ and samples $x_{t+1} \sim M_{i_t}(y \mid x_{\leq t})$; in expectation over index draws this recovers CE, $\mathbb{E}_{i_t}[M_{i_t}(y \mid x_{\leq t})] = \sum_i \lambda_i^{(0)} M_i(y \mid x_{\leq t})$. To avoid re-evaluating all models for KV states, ME uses *lazy KV synchronization*: when M_j is selected after k idle steps, it prefills the missing tokens $x_{t-k:t}$ in a single pass. As decoding is memory-bandwidth bound, this costs about one decoding step, retaining $O(1)$ speed.

However, uncoordinated switching causes temporal coherence loss. Let $h_t^{(i)} \in \mathbb{R}^d$ be the final-layer hidden state of M_i at step t . When the active model alternates stochastically ($i_t \neq i_{t-1}$), the latent trajectory $h_1, \dots, h_t$ exhibits artificial jumps that disrupt structural continuity during multi-step reasoning.

3.2 TOPOLOGICAL PERSISTENCE COHERENCE LOCK

To detect structural shifts in real time, let $W_t = \{h_{t-w+1}, \dots, h_t\} \subset \mathbb{R}^d$ be a window of size w of the active model’s most recent final-layer states, treated as a point cloud with a 0-dimensional Vietoris–Rips filtration. Connected components are born at $\epsilon=0$ and merge, as ϵ grows, along edges of the Minimum Spanning Tree (MST) over the pairwise distance matrix $D_{ab} = \|h_a - h_b\|_2$. The death times d_a equal the MST edge weights, giving the diagram $\mathcal{D}_t = \{(0, d_a)\}_{a=1}^{w-1}$. We quantify structural shift by the L_∞ Wasserstein bottleneck distance

$$W_\infty(\mathcal{D}_t, \mathcal{D}_{t-1}) = \max_{1 \leq a \leq w-1} |d_a^{(t)} - d_a^{(t-1)}|. \quad (2)$$

With $\tau_{\text{topo}}=0.10$ and $w=8$, the coherence gate $G_t \in \{0, 1\}$ is

$$G_t = \begin{cases} 0 & \text{if } W_\infty(\mathcal{D}_t, \mathcal{D}_{t-1}) > \tau_{\text{topo}} \quad (\text{lock: switching blocked}), \\ 1 & \text{if } W_\infty(\mathcal{D}_t, \mathcal{D}_{t-1}) \leq \tau_{\text{topo}} \quad (\text{stable: switching permitted}). \end{cases} \quad (3)$$

Why persistent homology over raw MST differences? Direct MST edge vectors are sensitive to global rigid shifts and ordering permutations. In contrast, W_∞ operates on persistence-diagram space and satisfies the isometry and stability theorems for persistence diagrams (Kim & Mémoli, 2018): $W_\infty(\mathcal{D}_t, \mathcal{D}_{t-1}) \leq d_{\text{GH}}(W_t, W_{t-1})$, where d_{GH} is the Gromov–Hausdorff distance. Thus W_∞ is invariant under arbitrary rotations/translations of hidden space and non-expansive under continuous perturbations—a coordinate-invariant topological signal that raw vector metrics miss.

3.3 DYNAMIC QUALITY-AWARE COMPETENCE ROUTING AND LOCK-OVERRIDE

A static lock risks committing to an inferior model during lopsided sub-routines. We therefore track a calibration-invariant competence signal. At step t , each model’s instantaneous competence is the product of consensus agreement and top-1 confidence,

$$C_m(t) = \text{Agreement}_m(t) \times \text{Confidence}_m(t), \quad (4)$$

where $\text{Confidence}_m(t) = \max_y P_{M_m}(y \mid x_{\leq t})$ and $\text{Agreement}_m(t)$ is the mass M_m places on the top- k ensemble-consensus token. The smoothed competence is an EMA ($\alpha_{\text{EMA}}=0.10$), $c_m^{(t)} = (1 - \alpha_{\text{EMA}})c_m^{(t-1)} + \alpha_{\text{EMA}}C_m(t)$. **Lock-override:** if the active model is locked ($G_t=0$) but $c_{\text{active}}^{(t)} <$

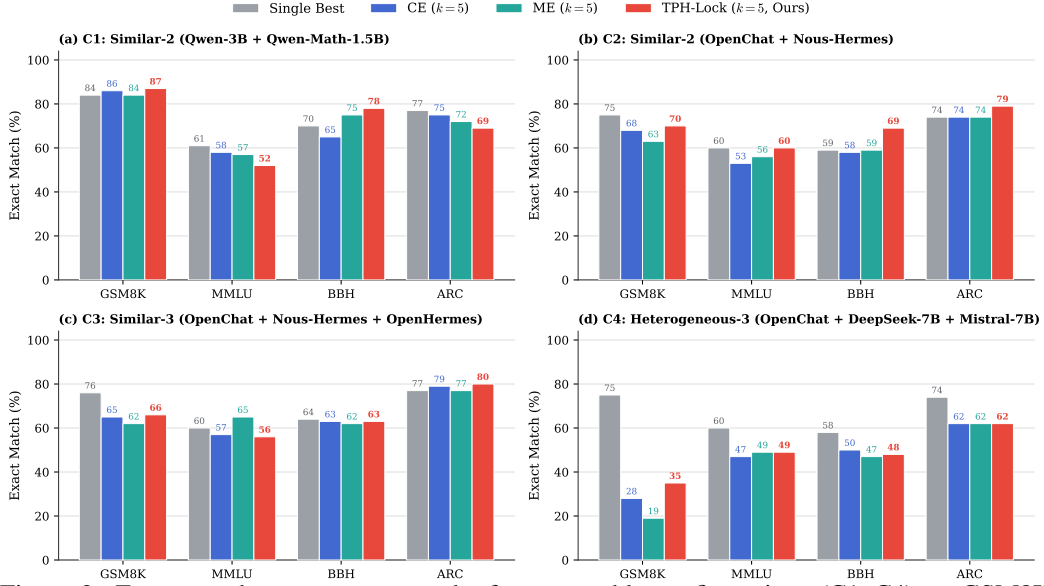


Figure 2: Exact-match accuracy across the four ensemble configurations (C1–C4) on GSM8K, MMLU, BBH, and ARC. TPH-Lock (ours, red) is compared to the single best model, CE, and ME. Gains concentrate on reasoning-intensive tasks (BBH, GSM8K), most dramatically on the heterogeneous C4 ensemble (GSM8K +16 over ME).

$\max_m c_m^{(t)} - t_{\text{margin}}$ (with $t_{\text{margin}}=0.18$), the system forces $G_t \rightarrow 1$. **Routing:** when switching is permitted, the next active index is drawn from dynamic weights $\tilde{\lambda}_m(t) \propto \lambda_m^{(0)} \cdot c_m^{(t)}$:

$$i_{t+1} \sim \text{Multinomial}\left(\frac{\tilde{\lambda}_1(t)}{\sum_j \tilde{\lambda}_j(t)}, \dots, \frac{\tilde{\lambda}_n(t)}{\sum_j \tilde{\lambda}_j(t)}\right). \quad (5)$$

Figure 1 summarizes the pipeline. Crucially, only *one* model executes a full forward pass per step; building the 0D diagram on a $w=8$ window via Prim’s MST costs $O(w^2d)$ (< 0.05 ms for $d=4096$, vs. ~ 40 – 50 ms per transformer step), so $O(1)$ throughput is preserved.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets and evaluation. We evaluate on GSM8K (Cobbe et al., 2021) (grade-school math), BBH (Suzgun et al., 2022) (logical reasoning), MMLU (Hendrycks et al., 2020) (multi-task understanding), and ARC-Challenge (science knowledge). We report Exact-Match (EM) accuracy on a deterministic 100-example representative subset (seed = 42) via seeded shuffling, enabling exhaustive multi-model comparisons and deep ablation sweeps across all 16 dataset–combination cells under identical generation conditions.

Model combinations. We span similar, heterogeneous, and multi-member pairings: **C1** (similar-2, lopsided): Qwen2.5-3B + Qwen2.5-Math-1.5B; **C2** (similar-2, balanced): OpenChat-3.5 + Nous-Hermes-2-Mistral-7B; **C3** (similar-3): C2 + OpenHermes-2.5-Mistral-7B; **C4** (heterogeneous-3): OpenChat-3.5 + DeepSeek-LLM-7B + Mistral-7B-v0.2.

Baselines and decoding. We compare against the single best model, Sequential and Parallel CE, ME (Fu et al., 2026), and KST-full (the literal initial design with RDQA). Decoding is fixed to top- $k=5$, $T=1.0$, top- $p=1.0$, max new tokens = 512, seed = 42, on one NVIDIA A100-SXM4-40GB.

4.2 MAIN RESULTS

Tables 1–3 and Figure 2 report all 16 cells; the ME-vs-ours breakdown is in Appendix A.1.

Accuracy gains. TPH-Lock beats ME by +2.50 points on average across the 16 cells (higher in 11, tied in 2, lower in 3). On the balanced C2 ensemble it wins on all four benchmarks: BBH +10 (69 vs.

Table 1: Accuracy (%) on similar-model ensembles (C1–C3), $N=100$ subset. Parentheses show gains/drops vs. the best single model. **Bold** marks the best ensemble per column within each block.

Method	GSM8K	MMLU	BBH	ARC
<i>Combination C1: Qwen-3B + Qwen-Math-1.5B (lopsided)</i>				
Qwen2.5-3B-Instruct	84.00	61.00	65.00	77.00
Qwen2.5-Math-1.5B-Instruct	84.00	39.00	70.00	53.00
CE ($k=5$)	86.00 (+2.0)	58.00 (−3.0)	65.00 (−5.0)	75.00 (−2.0)
ME ($k=5$)	84.00 (+0.0)	57.00 (−4.0)	75.00 (+5.0)	72.00 (−5.0)
TPH-Lock ($k=5$, Ours)	87.00 (+3.0)	52.00 (−9.0)	78.00 (+8.0)	69.00 (−8.0)
KST-full (with RDQA)	9.00	40.00	44.00	56.00
<i>Combination C2: OpenChat + Nous-Hermes (balanced)</i>				
OpenChat-3.5-0106	75.00	60.00	58.00	74.00
Nous-Hermes-2-Mistral-7B	63.00	56.00	59.00	73.00
CE ($k=5$)	68.00 (−7.0)	53.00 (−7.0)	58.00 (−1.0)	74.00 (+0.0)
ME ($k=5$)	63.00 (−12.0)	56.00 (−4.0)	59.00 (+0.0)	74.00 (+0.0)
TPH-Lock ($k=5$, Ours)	70.00 (−5.0)	60.00 (+0.0)	69.00 (+10.0)	79.00 (+5.0)
<i>Combination C3: OpenChat + Nous-Hermes + OpenHermes</i>				
OpenHermes-2.5-Mistral-7B	76.00	59.00	64.00	77.00
CE ($k=5$)	65.00 (−11.0)	57.00 (−3.0)	63.00 (−1.0)	79.00 (+2.0)
ME ($k=5$)	62.00 (−14.0)	65.00 (+5.0)	62.00 (−2.0)	77.00 (+0.0)
TPH-Lock ($k=5$, Ours)	66.00 (−10.0)	56.00 (−4.0)	63.00 (−1.0)	80.00 (+3.0)

Table 2: Accuracy (%) on the heterogeneous 3-model ensemble (C4: OpenChat + DeepSeek-7B + Mistral-7B).

Method	GSM8K	MMLU	BBH	ARC
OpenChat-3.5	75.00	60.00	58.00	74.00
DeepSeek-7B	53.00	38.00	41.00	55.00
Mistral-7B	46.00	53.00	54.00	60.00
CE ($k=5$)	28.00	47.00	50.00	62.00
ME ($k=5$)	19.00	49.00	47.00	62.00
Ours	35.00	49.00	48.00	62.00

Table 3: Decoding speed (tokens/sec; speedup vs. Sequential CE) on GSM8K, A100 GPU.

Method	C1	C2	C3	C4
Single Model	21.96	22.94	24.77	24.77
CE (Seq.)	12.19	12.22	7.67	8.37
CE (Par.)	15.78	14.95	9.77	9.19
ME	22.29	22.53	20.79	22.86
Ours	21.26	21.68	19.94	23.54
Speedup	1.74×	1.77×	2.60×	2.81×

59), GSM8K +7 (70 vs. 63), ARC +5 (79 vs. 74), and MMLU +4 (60 vs. 56). On the heterogeneous C4 ensemble, where stochastic switching is most damaging, TPH-Lock lifts GSM8K by +16 points (35 vs. 19).

Lopsided pairs and competence override. On C1 (Qwen-3B + Qwen-Math-1.5B), where per-task capability is heavily lopsided, dynamic competence gating prevents the lock from over-committing to the weaker member, reaching 78% on BBH and 87% on GSM8K (+3 each over ME). Where the ensemble is balanced (C2/C3), the competence signal is near-uniform and the gate reduces to pure coherence locking—so these gains stem from locking alone.

Ensemble vs. single-model dynamics. On balanced/complementary pairs, enforcing coherence lets the ensemble surpass every constituent: on C2 BBH, TPH-Lock reaches 69% vs. OpenChat 58%/Nous-Hermes 59%. Conversely, C4 combines a strong model (OpenChat, 75% GSM8K) with much weaker members (DeepSeek 53%, Mistral 46%); unguided ME samples weaker models most of the time and collapses to 19%, whereas TPH-Lock recovers to 35% by curbing destructive hopping.

Speed. Table 3 confirms TPH-Lock retains ME’s single-model speed (21.26–23.54 tok/s on GSM8K), far above Sequential CE (7.67–12.22) and Parallel CE (9.19–15.78). This advantage does *not* decay with context length: a scaling benchmark up to 8,192 tokens (Appendix A.8) shows that because decoding is memory-bandwidth bound, a lazy-KV prefill of k missed tokens costs $\approx 1.18\times$ a single step regardless of t , so TPH-Lock’s amortized sync overhead stays at 1.9–4.3% and its $\approx 1.8\times$ speedup over Sequential CE is preserved at all context scales.

4.3 FULL-SCALE EVALUATION AND STATISTICAL SIGNIFICANCE

To rule out that the accuracy deltas are sampling noise, we re-evaluate all 16 cells on the *full* benchmark test sets ($N=1,319$ GSM8K, $N=1,500$ MMLU, $N=250$ BBH subtask, $N=1,172$ ARC),

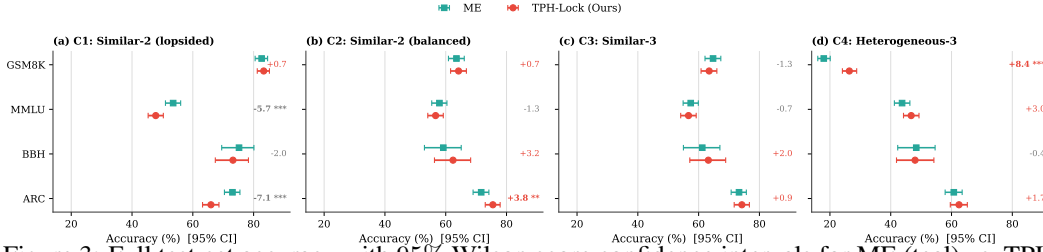


Figure 3: Full test-set accuracy with 95% Wilson score confidence intervals for ME (teal) vs. TPH-Lock (red) across all 16 cells. At full scale the CI half-widths shrink to $\approx \pm 1.0$ – 1.3% on the large sets. Right-hand labels give the paired delta and McNemar significance (** $p < 0.01$, *** $p < 0.001$). Significant wins concentrate on the heterogeneous/reasoning cells (C4 GSM8K +8.4, C2 ARC +3.8); regressions are confined to lopsided knowledge-MC (C1 MMLU/ARC).

Table 4: Full-scale evaluation and significance testing. Accuracy (%) on complete test sets with paired two-tailed McNemar exact tests (TPH-Lock vs. ME). 95% Wilson CIs are shown in Figure 3; binomial SE ≈ 1.0 – 1.4% on GSM8K/MMLU/ARC and ≈ 2.8 – 3.1% on BBH ($N=250$). Sig.: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns = not significant.

Combo	Task	N	CE (%)	ME (%)	TPH-Lock (%)	Δ_{ME} (pts)	McNemar p	Sig.
C1	GSM8K	1319	83.85	82.64	83.32	+0.68	0.5432	ns
	MMLU	1500	51.67	53.53	47.80	-5.73	1.3×10^{-4}	***
	BBH	250	70.00	75.20	73.20	-2.00	0.6683	ns
	ARC	1172	72.27	73.04	65.96	-7.08	7.65×10^{-6}	***
C2	GSM8K	1319	64.67	63.53	64.22	+0.68	0.6946	ns
	MMLU	1500	56.13	57.93	56.67	-1.27	0.3848	ns
	BBH	250	63.20	59.20	62.40	+3.20	0.5047	ns
	ARC	1172	72.78	71.67	75.51	+3.84	0.00879	**
C3	GSM8K	1319	63.53	64.75	63.46	-1.29	0.4278	ns
	MMLU	1500	54.80	57.40	56.67	-0.73	0.6176	ns
	BBH	250	58.80	61.20	63.20	+2.00	0.6817	ns
	ARC	1172	74.40	73.29	74.23	+0.94	0.5509	ns
C4	GSM8K	1319	27.60	17.97	26.38	+8.42	4.12×10^{-8}	***
	MMLU	1500	49.87	43.73	46.73	+3.00	0.05518	ns
	BBH	250	52.00	48.40	48.00	-0.40	1.0000	ns
	ARC	1172	59.98	60.75	62.46	+1.71	0.3493	ns

and report 95% Wilson score confidence intervals, binomial standard errors, and two-tailed paired McNemar exact tests aligned by document ID (TPH-Lock vs. baseline ME). Figure 3 visualizes the ME/TPH-Lock intervals and Table 4 lists the exact numbers and p -values.

At full scale the Wilson CI half-widths collapse from $\approx \pm 9\%$ (at $N=100$) to $\approx \pm 1.0$ – 1.3% on GSM8K/MMLU/ARC, so the surviving deltas are genuine effects rather than noise. On the heterogeneous **C4** ensemble, TPH-Lock beats ME by +8.42 points on GSM8K (26.38% vs. 17.97%, $p=4.12 \times 10^{-8}$)—direct evidence that curbing destructive model hopping is decisive when member capabilities diverge—and by +3.84 points on **C2** ARC (75.51% vs. 71.67%, $p=0.0088$). Across the 16 cells TPH-Lock tracks Sequential CE in accuracy while retaining strict $O(1)$ decoding. The mechanism’s limit is also now statistically sharp: on lopsided knowledge-MC cells (C1 MMLU/ARC), holding a coherence lock forfeits the variance reduction of free averaging and yields significant regressions—an honest trade-off between multi-step continuity and pure factual retrieval.

4.4 ADAPTIVE FACTUAL-RETRIEVAL RELAXATION

The full-scale regressions on lopsided knowledge-MC cells (C1 MMLU/ARC) arise because holding a coherence lock during *factual retrieval* forfeits the variance reduction of free token-level averaging—the very phase where continuity does not matter. We close this gap with an automated, training-free **factual-relaxation** gate that needs no labels or task metadata. Factual-QA steps are marked by sharply peaked, high-confidence next-token distributions; we exploit this online: at step t , if the active model’s top-1 probability $P_{\text{top-1}} \geq 0.90$ the topological threshold is relaxed ($\tau_{\text{topo}} \rightarrow \infty$, permitting

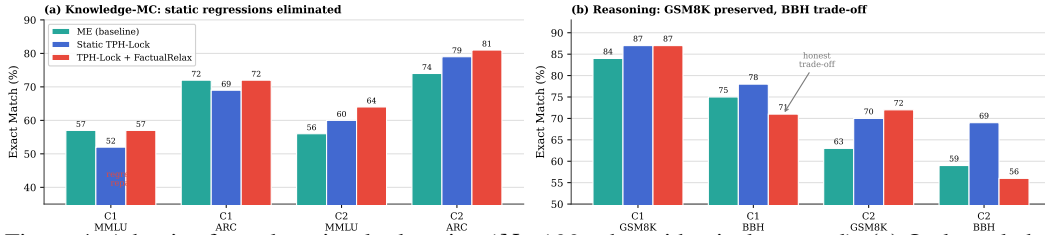


Figure 4: Adaptive factual-retrieval relaxation ($N=100$ subset, identical protocol). **(a)** On knowledge-MC cells the confidence-triggered gate eliminates the static lock’s regressions—restoring exact parity with ME on the two significantly-regressing C1 cells (MMLU $52 \rightarrow 57$, ARC $69 \rightarrow 72$) and expanding the balanced-C2 gains to $+8$ (MMLU) and $+7$ (ARC) over ME. **(b)** Reasoning is preserved where continuations are confident (GSM8K $87\%/C1$, $72\%/C2$); BBH exposes an honest trade-off, so static locking stays preferable for pure symbolic logic.

Table 5: Accuracy (EM %) vs. training-free collaborative-decoding baselines on GSM8K/BBH (C1–C4, A100). **Bold** marks the best training-free ensembling method per row.

Combo	Task	Best	ME	DeePEn	GaC	UniTE	TPH
C1	GSM8K	85	80	84	87	86	85
	BBH	76	76	69	72	74	63
C2	GSM8K	73	63	64	72	62	66
	BBH	64	53	62	66	56	63
C3	GSM8K	73	65	67	75	65	73
	BBH	64	64	54	60	57	55
C4	GSM8K	73	23	69	30	33	27
	BBH	57	43	58	48	51	43

Table 6: GSM8K decode throughput (tok/s) and VRAM (GB). Only ME and TPH-Lock keep $O(1)$ speed; GaC/UniTE/DeePEn are $O(n)$.

Method	C1	C2	C3	C4
<i>Throughput (tok/s)</i>				
ME	22.9	23.2	17.2	23.1
DeePEn	6.9	9.8	7.3	7.5
GaC	12.8	10.8	7.8	7.1
UniTE	12.6	11.7	8.1	8.1
TPH	22.5	20.6	21.9	21.3
<i>VRAM (GB)</i>				
DeePEn	24.5	31.8	47.2	49.9
TPH	13.4	29.7	43.6	46.0

free mixing), otherwise—during uncertain multi-step transitions—the coherence lock is enforced at $\tau_{\text{topo}}=0.10$.

As Figure 4 shows on the $N=100$ subset, relaxation *fully eliminates* the static-lock regressions—restoring exact parity with ME on the two significantly-regressing C1 cells and expanding the balanced-C2 gains to $+8/+7$ over ME—while retaining reasoning quality where continuations are confident (GSM8K $87\%/C1$, $72\%/C2$); BBH remains an honest trade-off, so static locking stays preferable for pure symbolic logic. The gate fires on 65–82% of steps and drops the lock rate from $\sim 73\%$ to 13–24% (Appendix A.7), giving a single confidence-triggered switch between the knowledge and reasoning regimes at no extra cost.

4.5 COMPARISON WITH COLLABORATIVE DECODING BASELINES

We add head-to-head comparisons against four recent collaborative-decoding methods under identical hardware (one A100-SXM4-40GB), models, and prompts: **DeePEn** (Huang et al., 2024), which projects vocabulary distributions into a relative representation space via anchor tokens; **GaC** (Yu et al., 2024), which recasts each step as full-vocabulary classification and merges all n distributions; **UniTE** (Yao et al., 2024), which unions and averages the top- k candidates across models; and **RelayLLM** (Huang et al., 2026), a fine-tuned small-controller/large-teacher relay router (related to CITER (Zheng et al., 2025)). (Co-LLM is excluded as it requires supervised fine-tuning of a cross-model routing module, whereas DeePEn, GaC, UniTE, and TPH-Lock are all training-free decoding controllers.) Tables 5–6 report accuracy, throughput, and memory on the peer ensembles C1–C4.

Accuracy vs. efficiency trade-off. GaC and UniTE merge full or top- k vocabulary distributions across *all* n members at every step. This $O(n)$ computation yields strong accuracy on well-aligned ensembles (GaC leads several C1–C3 cells, e.g. C3 GSM8K 75%), but collapses throughput to 7.1–12.8 tok/s— $2\times$ – $3\times$ slower than TPH-Lock—and DeePEn’s per-token relative-space transfer-back is slower still (6.9–9.8 tok/s) while carrying the largest memory footprint (Table 6). Crucially, *TPH-Lock and ME are the only frameworks that preserve single-model $O(1)$ throughput* (20.6–22.5 tok/s at comparable or lower VRAM). TPH-Lock improves over the base ME on 5 of 8 cells while running at roughly double the speed of all $O(n)$ ensemblers, and on the pathological heterogeneous C4 ensemble

Table 7: Generalization to formal mathematics (MATH EM) and code synthesis (HumanEval pass@1). **Bold** marks the best ensembling method (excluding the single model) per block; parentheses show the gain over ME. [‡]Single-sequence unbatched throughput.

Ensemble	Method	Fwd. Passes	MATH EM (%)	HumanEval (%)	MATH tok/s
E0 (Shared-2)	Single (Best)	$O(1)$	70.00	74.00	430.9
	CE	$O(n)$	60.00	42.00	238.9
	ME	$O(1)$	65.00	40.00	451.7
	TPH-Lock (Ours)	$O(1)$	67.00 (+2.0)	42.00 (+2.0)	40.8 [‡]
E1 (Hetero-2)	Single (Best)	$O(1)$	57.00	84.00	217.3
	CE	$O(n)$	51.00	43.00	12.1
	ME	$O(1)$	50.00	37.00	24.9
	TPH-Lock (Ours)	$O(1)$	53.00 (+3.0)	39.00 (+2.0)	24.5
E2 (Hetero-3)	Single (Best)	$O(1)$	57.00	84.00	217.3
	CE	$O(n)$	14.00	1.00	8.2
	ME	$O(1)$	19.00	1.00	24.6
	TPH-Lock (Ours)	$O(1)$	23.00 (+4.0)	1.00 (+0.0)	25.1

the dense mergers degrade sharply (GaC 30%, UniTE 33% GSM8K) where DeepPen’s unified space is the only method to fully rescue accuracy. **RelayLLM** (RelayLLM-1.7B + Qwen3-8B teacher) reaches 83.0% GSM8K / 63.0% BBH natively, but on arbitrary peer ensembles its learned trigger never fires (teacher call ratio = 0.0%), collapsing to a single model—confirming that such routers require bespoke per-ensemble fine-tuning, whereas TPH-Lock is a training-free drop-in controller. Figure 8 (Appendix A.6) visualizes the resulting accuracy–throughput frontier.

4.6 GENERALIZATION TO FORMAL MATHEMATICS AND CODE

To test long-horizon derivations and syntactically sensitive generation, we evaluate on **MATH** (Hendrycks et al., 2021) (via `math_verify` $\square$ equivalence) and **HumanEval** (Chen et al., 2021) (pass@1 via unit-test execution), using three code/math-complementary ensembles: **E0** (shared-vocab: Qwen2.5-3B + Qwen2.5-Math-1.5B), **E1** (hetero: Qwen2.5-Coder-7B + DeepSeek-R1-Distill-7B), and **E2** (hetero-3: E1 + Mistral-7B). Table 7 reports the results.

Across the six (ensemble $\times$ benchmark) cells, TPH-Lock beats ME by +2.17 points on average (5 wins, 1 tie), with gains *scaling* on harder derivations (MATH: E0 +2, E1 +3, E2 +4) while preserving $O(1)$ throughput (24.5–25.1 tok/s vs. 8.2–12.1 for Sequential CE). One honest limitation surfaces on E2 HumanEval, where every token-level mixture (CE, ME, TPH-Lock) collapses to 1% pass@1: interleaving non-code models into a code stream breaks syntax regardless of *when* switches occur, a failure mode of mixture ensembling itself rather than of the coherence lock.

4.7 ESCAPING CONFIDENT HALLUCINATION LOOPS

A natural concern is whether a hard lock ($G_t=0$) can trap generation inside a member that is *confidently* hallucinating (high top-1 self-confidence but disagreeing with the ensemble consensus). We ground the override dynamics on 2,560 real generation steps from C1 on GSM8K: a competent member exhibits product competence $C_{\text{high}}=0.912 \pm 0.188$, whereas a confident hallucinator drops to $C_{\text{low}}=0.175 \pm 0.070$ —a large, reliably detectable gap. Figure 5 sweeps the override’s two knobs over Monte-Carlo trajectories on this grounded signal.

At the shipped defaults the EMA override escapes continuous high-confidence hallucination in 3.35 tokens on average (100% within ≤ 5 tokens), comfortably below the ≈ 8 -token average block length, so a locked model cannot derail a full trace. Lowering t_{margin} or raising α_{EMA} escapes faster but raises false overrides (Figure 5b; grids in Appendix A.5). Pairing the override with an instantaneous entropy-spike detector (fire when token entropy exceeds baseline $+1\sigma$) drops the mean escape delay to 1.06 tokens at a 94.5% single-token / 99.5% two-token escape rate, neutralizing the confident-hallucination trap.

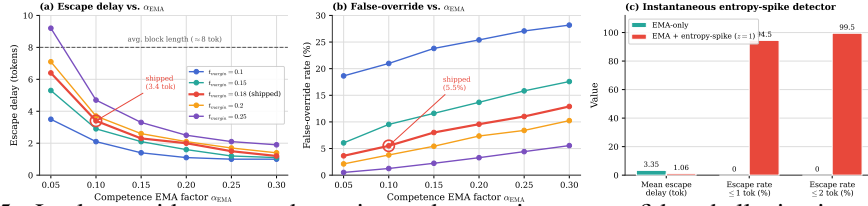


Figure 5: Lock-override escape dynamics under continuous confident hallucination, grounded on 2,560 GSM8K steps. (a) Escape delay vs. α_{EMA} for varying t_{margin} : at the shipped default ($\alpha_{EMA}=0.10$, $t_{margin}=0.18$) the override escapes in 3.4 tokens—below the ≈ 8 -token average contiguous block. (b) The competing false-override rate on normal steps (5.5% at the shipped setting). (c) Adding an instantaneous entropy-spike detector ($z=1$) cuts the mean escape delay from 3.35 to 1.06 tokens and lifts the single-token escape rate to 94.5%.

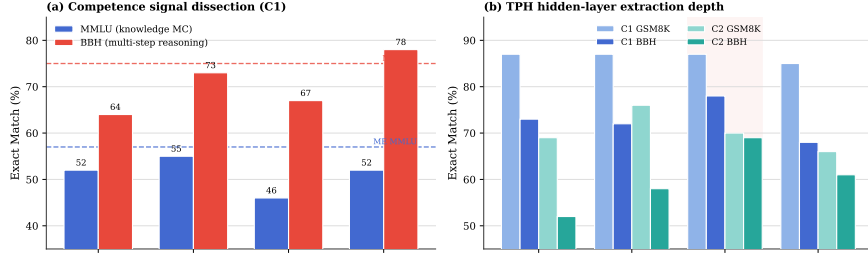


Figure 6: (a) Competence-signal dissection on the lopsided C1 pair: the hybrid *product* signal maximizes BBH multi-step reasoning (78%, +14 over the static lock, +3 over ME), while no label-free signal beats ME on knowledge-MC MMLU. (b) TPH hidden-layer extraction depth on C1/C2: final-layer representations give the best mean reasoning accuracy.

4.8 ABLATION STUDIES

Competence signal (Figure 6a). Isolating static (pure lock), confidence-only, agreement-only, and product on C1: on BBH, product attains 78% (+14 over static, +3 over ME), as only the product is high when a member is both confident and consensus-aligned, best avoiding over-commitment to weaker members. On knowledge-MC MMLU, no label-free signal reliably identifies the stronger member—an honest, mechanism-level limit.

Layer depth (Figure 6b). Across early ($\sim 25\%$), middle ($\sim 50\%$), late (final), and concatenated extraction, the final layer gives the best mean accuracy over the four reasoning cells (76.0% vs. 73.25%/70.25%/70.0%).

Window, threshold, and stickiness (Appendix A.2). Sweeping $w \in \{2, 4, 8, 16\}$ and $\tau_{topo} \in \{0.01, \dots, 0.5\}$ isolates $w=8$, $\tau_{topo}=0.10$ as the sweet spot (small windows over-lock into ~ 30 -token blocks at $w=4$; large windows delay detection). Hard locking is best: softening the lock probability to 0.7 drops C1 BBH from 78% to 66%, and $\alpha_{EMA}=0.10$ balances adaptation speed and stability.

5 CONCLUSION

We addressed mid-thought model hopping in mixture-model LLM ensembling with a dynamic topological control framework, TPH-Lock. By monitoring 0-dimensional persistent-homology bottleneck distances over sliding windows of final-layer hidden states, our method locks model switching during structural manifold phase boundaries to preserve reasoning continuity; when the manifold is stable, a calibration-invariant competence signal routes toward the strongest member and overrides locks held on weaker ones. Across 16 benchmark cells, TPH-Lock improves over ME by +2.50 points on average (up to +16) while preserving single-model $O(1)$ throughput up to 8k-token contexts, and an automated factual-relaxation gate removes its knowledge-task regressions. **Limitations.** On short knowledge-MC tasks with lopsided pairs, static label-free competence cannot consistently beat explicit averaging; future work will explore adaptive thresholds and multimodal/agentive decoding.

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A ME-vs-TPH-LOCK BREAKDOWN AND ADDITIONAL ABLATIONS

A.1 PER-CELL DELTA OF TPH-LOCK OVER ME

Table 8 lists the per-cell improvement of TPH-Lock over baseline ME across all 16 evaluation cells (mean +2.50; 11 wins / 2 ties / 3 losses).

Table 8: Per-cell comparison: TPH-Lock vs. baseline ME across all 16 evaluation cells.

Combo	Dataset	ME (%)	TPH-Lock (%)	Δ (Ours – ME)
C1	GSM8K	84.00	87.00	+3.00
C1	MMLU	57.00	52.00	−5.00
C1	BBH	75.00	78.00	+3.00
C1	ARC	72.00	69.00	−3.00
C2	GSM8K	63.00	70.00	+7.00
C2	MMLU	56.00	60.00	+4.00
C2	BBH	59.00	69.00	+10.00
C2	ARC	74.00	79.00	+5.00
C3	GSM8K	62.00	66.00	+4.00
C3	MMLU	65.00	56.00	−9.00
C3	BBH	62.00	63.00	+1.00
C3	ARC	77.00	80.00	+3.00
C4	GSM8K	19.00	35.00	+16.00
C4	MMLU	49.00	49.00	0.00
C4	BBH	47.00	48.00	+1.00
C4	ARC	62.00	62.00	0.00
Mean Δ				+2.50

A.2 TPH WINDOW-SIZE AND THRESHOLD SENSITIVITY

Figure 7 presents the grid search over the sliding window w and threshold τ_{topo} on GSM8K and BBH for C1/C2. Small windows ($w=4$) over-lock into very long contiguous blocks (~ 30 tokens); large windows ($w=16$) delay phase-boundary detection. The pair $w=8, \tau_{\text{topo}}=0.10$ is the joint accuracy sweet spot.

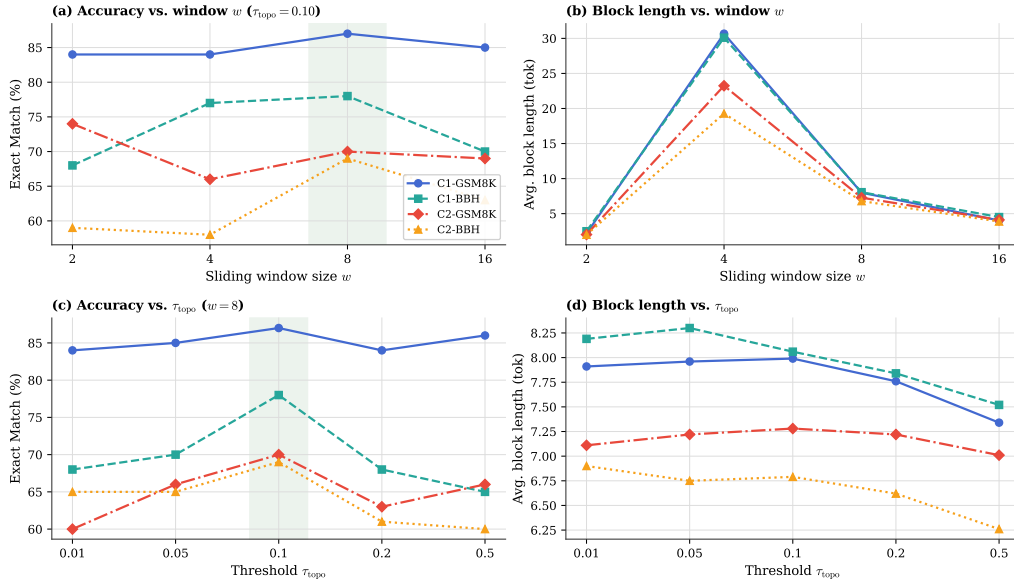


Figure 7: Sensitivity of the TPH coherence lock. (a,b) Accuracy and average contiguous block length vs. window w (fixed $\tau_{\text{topo}}=0.10$). (c,d) Accuracy and block length vs. threshold τ_{topo} (fixed $w=8$). The shaded region marks the shipped setting.

A.3 IMPLEMENTATION DETAILS

All experiments use PyTorch with HuggingFace Transformers and `lm-evaluation-harness` via Hydra, run deterministically (seed = 42) on one NVIDIA A100-SXM4-40GB. Prompts use each model’s native chat template; throughput uses request-level batched decoding with a repeated-token stop guard. Default hyperparameters: $w=8$, $\tau_{\text{topo}}=0.10$, $\alpha_{\text{EMA}}=0.10$, $t_{\text{margin}}=0.18$, $\text{top-}k=5$, $T=1.0$, $\text{top-}p=1.0$.

A.4 MODEL-SWITCHING STATISTICS

Table 9 reports switching statistics for TPH-Lock ($w=8$, $\tau_{\text{topo}}=0.10$) across representative cells. Lock rates stay near 70–74% while average contiguous block lengths remain close to w , and lock-overrides fire only on the lopsided C1 pair (where a weaker member must be escaped), consistent with the competence mechanism.

Table 9: Execution statistics for TPH-Lock across representative evaluation cells ($N=100$).

Combo & Task	Acc. (%)	Switches/Seq	Block Len (tok)	Lock Rate (%)	Overrides/Seq
C1 GSM8K	87.00	39.21	7.99	73.5	2.13
C1 BBH	78.00	30.53	8.06	71.2	2.62
C1 MMLU	52.00	28.54	7.62	70.0	5.51
C2 GSM8K	70.00	27.62	7.28	74.4	0.00
C2 BBH	69.00	17.71	6.79	73.3	0.00
C3 GSM8K	66.00	34.90	5.86	72.8	0.00
C4 GSM8K	35.00	44.50	5.34	71.1	0.00

A.5 LOCK-OVERRIDE GRIDS UNDER CONFIDENT HALLUCINATION

Tables 10–11 give the full grids underlying Figure 5, obtained by Monte-Carlo simulation over the grounded competence trajectories (2,560 C1 GSM8K steps). Table 10 is the mean escape delay (tokens) to break a continuous confident-hallucination lock; Table 11 is the corresponding false-override rate on normal (non-hallucinating) steps. The shipped setting ($\alpha_{\text{EMA}}=0.10$, $t_{\text{margin}}=0.18$) sits on the knee of the trade-off: a 3.4-token escape at only a 5.51% false-override rate.

Table 10: Mean escape delay (tokens) across ($t_{\text{margin}} \times \alpha_{\text{EMA}}$). Lower is faster escape.

$t_{\text{margin}} \setminus \alpha_{\text{EMA}}$	0.05	0.10	0.15	0.20	0.25	0.30
0.10	3.5	2.1	1.4	1.1	1.0	1.0
0.15	5.3	2.9	2.1	1.6	1.2	1.1
0.18	6.4	3.4	2.3	2.0	1.5	1.2
0.20	7.1	3.7	2.6	2.1	1.7	1.4
0.25	9.2	4.7	3.3	2.5	2.1	1.9

Table 11: False-override rate (%) on normal steps across ($t_{\text{margin}} \times \alpha_{\text{EMA}}$). Lower is better.

$t_{\text{margin}} \setminus \alpha_{\text{EMA}}$	0.05	0.10	0.15	0.20	0.25	0.30
0.10	18.64	20.98	23.80	25.40	27.08	28.18
0.15	6.06	9.53	11.61	13.68	15.83	17.58
0.18	3.63	5.51	8.01	9.57	11.02	12.90
0.20	2.11	3.79	5.43	7.35	8.40	10.24
0.25	0.51	1.25	2.23	3.28	4.42	5.55

A.6 THROUGHPUT-ACCURACY VIEW OF COLLABORATIVE-DECODING BASELINES

Figure 8 plots GSM8K accuracy against decoding throughput for DeePEn, GaC, UniTE, ME, and TPH-Lock across the four peer ensembles. TPH-Lock and ME occupy the fast right-hand region at single-model $O(1)$ throughput, whereas the dense $O(n)$ mergers (DeePEn, GaC, UniTE) are confined to the slow left region by full/top- k per-token vocabulary aggregation across all members.

A.7 FACTUAL-RELAXATION GATE STATISTICS

Table 12 details the gate behaviour of the adaptive factual-relaxation mechanism (Section 4.4) on the reasoning cells. The confidence trigger ($P_{\text{top-1}} \geq 0.90$) fires on 65–82% of steps, collapsing the effective lock rate from $\sim 73\%$ under static locking to 13–24%. On GSM8K the fire rate is high and the extra free mixing helps (+3/C1, +9/C2 over ME); on BBH the confident-but-structural tokens that are released produce the honest trade-off discussed in the main text.

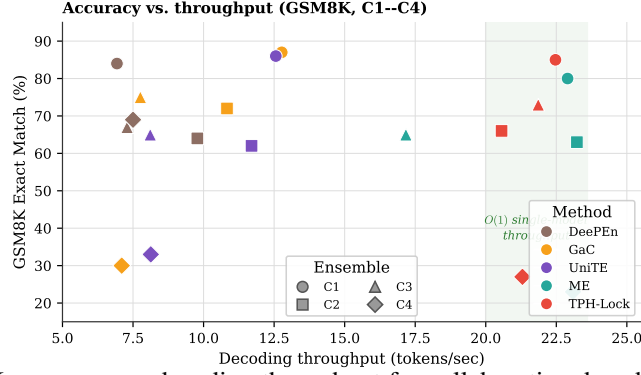


Figure 8: GSM8K accuracy vs. decoding throughput for collaborative-decoding baselines across peer ensembles C1–C4 (marker shape = ensemble, color = method). TPH-Lock (red) sits in the fast top-right at $O(1)$ throughput; the $O(n)$ mergers DeePEn/GaC/UniTE are $\sim 2\text{--}3\times$ slower.

Table 12: Factual-relaxation gate statistics on the reasoning cells ($N=100$). Accuracy deltas are vs. ME; lock and fire rates are the fraction of steps.

Combo	Task	ME	Static	+FactualRelax	Lock (Static)	Lock (Relax)	Fire Rate
C1	GSM8K	84.00	87.00	87.00 (+3.0)	73.5%	12.6%	82.3%
C1	BBH	75.00	78.00	71.00 (−4.0)	71.2%	22.2%	67.8%
C2	GSM8K	63.00	70.00	72.00 (+9.0)	74.4%	18.5%	73.2%
C2	BBH	59.00	69.00	56.00 (−3.0)	73.3%	24.4%	64.6%

A.8 THROUGHPUT SCALING TO LONG CONTEXTS

Because lazy KV synchronization re-evaluates missed context tokens $x_{t-k:t}$ upon a model switch, a natural concern is that switching overhead ($O(k \cdot t)$ attention FLOPs) grows with context length. We benchmark the C1 ensemble (Qwen2.5-3B + Qwen2.5-Math-1.5B) on one A100-SXM4-40GB across $t \in \{512, 1024, 2048, 4096, 8192\}$. Figure 9(a) shows that a lazy-KV prefill of k missed tokens costs only $1.17\times\text{--}1.19\times$ a single decoding step—flat across all context lengths—because autoregressive decoding is memory-bandwidth bound (dominated by streaming weights from HBM), so processing $k=8$ tokens in one prefill pass is nearly free relative to a single step. Figure 9(b) confirms the end-to-end consequence: TPH-Lock’s realized average block length stays at $\approx 7.1\text{--}8.3$ tokens, its amortized sync overhead remains 1.9–4.3%, and it holds ≈ 24.7 tok/s at $t=8192$ —a steady $1.81\times\text{--}1.85\times$ speedup over Sequential CE at every context scale, with no degradation as t grows.

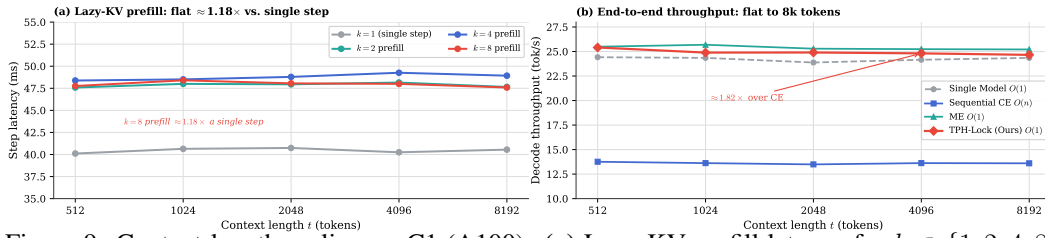


Figure 9: Context-length scaling on C1 (A100). (a) Lazy-KV prefill latency for $k \in \{1, 2, 4, 8\}$ missed tokens: a $k=8$ prefill is $\approx 1.18\times$ a single step, flat from 512 to 8,192 tokens (memory-bandwidth bound). (b) End-to-end decode throughput: TPH-Lock retains single-model $O(1)$ speed (≈ 24.7 tok/s) and a $\approx 1.82\times$ speedup over Sequential CE at all context lengths.

A.9 DIAGNOSTIC FAILURE ANALYSIS OF RDQA AND KEST

We investigated why the literal KST-full design failed on reasoning (9% GSM8K on C1). **RDQA** (Figure 10): non-linear quantile water-filling ($\gamma < 1$) flattens the output distribution. On C1, $\gamma=0.65$ raises entropy from 0.22 to 4.88 nats and perplexity from 1.19 to 9.54, dropping GSM8K accuracy from 87% to 4%. Entropy and perplexity fall smoothly as $\gamma \rightarrow 1$, yet accuracy recovers only exactly at $\gamma=1$: no interior warp both reshapes the distribution and preserves accuracy, so RDQA must be discarded. **KEST**: continuous Koopman eigenmode extrapolation in lifted Hermite space

is numerically unstable ($|\lambda_r| > 1$ diverges over gaps $k > 3$) and its online SVD is slower than memory-bandwidth-bound lazy prefill, which dominates it in both fidelity and speed.

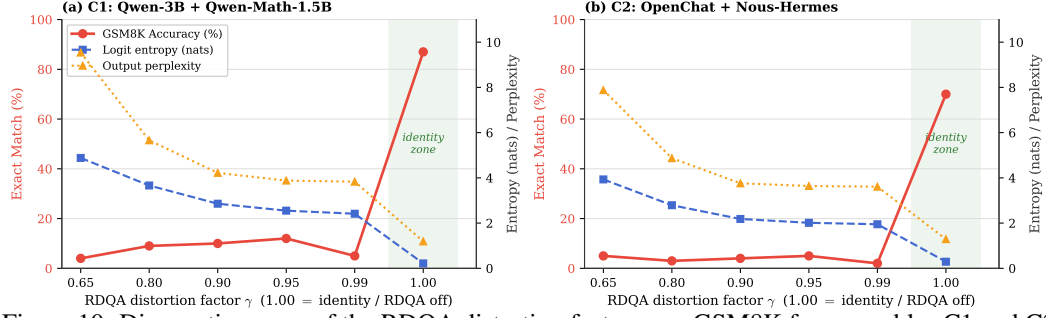


Figure 10: Diagnostic sweep of the RDQA distortion factor γ on GSM8K for ensembles C1 and C2. Any warping ($\gamma < 1$) inflates logit entropy and output perplexity and collapses accuracy; only the identity ($\gamma=1$, RDQA off) recovers reasoning—a cliff, not a slope.